



Ricardo Rey-Sáez

PSYCHOMETRICS · EXPERIMENTAL PSYCHOLOGY · BAYESIAN MODELING

PH.D. CANDIDATE

Universidad Autónoma de Madrid

Member of the [Cognition, Attention and Learning Lab](#) at Universidad Autónoma de Madrid (UAM). My research sits somewhere between Bayesian modelling, psychometrics, and experimental psychology. More specifically, I am interested in developing measurement models that can capture individual differences in how people perform on experimental tasks. For more information, visit my [personal website](#).

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Education

Ph.D. Psychology

2024–present

Universidad Autónoma de Madrid

- Developing and validating psychometric models for the study of individual differences in experimental psychology.

M.S. Quantitative Research Methods for Behavioral and Health Sciences

2020–2022

Complutense University of Madrid

B.S. Psychology

2017–2020

Complutense University of Madrid

International Research Visits

Visiting Ph.D. Researcher

Mar–May 2026

Universiteit van Amsterdam

- Supervised by Dr. Dora Matzke (see the [Amsterdam Mathematical Psychology Lab](#))

Additional Training

Measuring Individual Differences in Experimental Psychology (2.5 hours)

2026

XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS)

How do I preregister my study? A hands-on workshop (2.5 hours)

2026

XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS)

Bayesian Cognitive Modeling: A Practical Course

2024

Universiteit van Amsterdam

Practical Introduction to Advances in Exploratory Factor Analysis Applied to Item Analysis

2023

University of the Basque Country

Introduction to Exploratory Graph Analysis (EGA) with R

2022

Zaragoza University



Principles, Statistical and Computational Tools for Reproducible Data Science

2020

Harvard University (HarvardX) | [Verified Certificate](#)

Professional Certificate in Data Science

2020

Harvard University

- [Data Science: R basics](#) | [Verified Certificate](#)
- [Data Science: Visualization](#) | [Verified Certificate](#)
- [Data Science: Probability](#) | [Verified Certificate](#)
- [Data Science: Inference and Modeling](#) | [Verified Certificate](#)
- [Data Science: Productivity Tools](#) | [Verified Certificate](#)
- [Data Science: Wrangling](#) | [Verified Certificate](#)
- [Data Science: Linear Regression](#) | [Verified Certificate](#)

Teaching and Educational Activities

Meta-analysis, Latent Class and Latent Profile Analysis

2025

Lead instructor, Advanced Statistical Training for Researchers

- Organized by Deusto FamilyPsych, Deusto University

Introduction to Bayesian Psychometrics: Applications to Factor Analysis

2025

Seminar instructor, National interfaculty training seminar

- Organized by ASCOFAPSI (Colombian Association of Faculties of Psychology)

Beyond Maximum Likelihood: Introduction to Bayesian Factor Analysis

2025

Seminar instructor, Master's program in Quantitative Research Methods

- Organized by Autonomous University of Madrid

ChatGPT and Psychometrics: Optimizing the Design, Refinement, and Data Analysis of Measurement Instruments

2024

Seminar instructor, Master's program in Quantitative Research Methods

- Organized by Autonomous University of Madrid

Data Analysis in Psychology

2023–2024

Course instructor, Undergraduate program (Bachelor's in Psychology)

- Organized by Nebrija University

Adapting Our Teaching to the Revolution of AI and ChatGPT

2023

Workshop instructor, Faculty training

- Organized by Universidad Autónoma de Madrid

How can we Adapt Our Teaching to the Revolution of AI and ChatGPT?

2023

Workshop instructor, Faculty training

- Organized by University of Costa Rica

Textbooks

Rey-Sáez, R., & Estrada, E. (2023). *Análisis de datos con R en Ciencias Sociales y de la Salud*. Editorial Síntesis: <https://www.sintesis.com/libro/analisis-de-datos-con-r-en-ciencias-sociales-y-de-la-salud>

Software

BayesianEFA. R package for Bayesian exploratory factor analysis. Source code: <https://github.com/RicardoReySaez/BayesianEFA>

uSDT. R package for hierarchical signal detection theory models in unconscious processing research. Co-developed with F. Garre-Frutos, A. Franco-Martínez, and M. A. Vadillo. Source code: <https://github.com/RicardoReySaez/uSDT>

Awards

Best Oral Communication Award

2025

Ibero-American Association for Research on Individual Differences

- Presentation Title: **New Hierarchical Models to Answer Old Psychometric Questions in Experimental Research: Which Task Best Measures Inhibition?**
- Prize: **MASK-5** Measurement Instrument

Dissemination and Transfer in Methodology Award

2024

Spanish Association of Methodology for Behavioral Sciences

- Project **Ecuación Triunfo**
- Funding: €1,000

Julio Olea Grant

2024

Spanish Association of Methodology for Behavioral Sciences

- Funding to carry out the research project **The union of psychometrics and experimental psychology: Bayesian hierarchical models with common factors for response times**
- Funding: €2,400

Publications

1. **Rey-Sáez, R.**, & Revuelta, J. (2026). An Efficient Rotation-Sign-Permutation Algorithm to Solve Rotational Indeterminacy in Bayesian Exploratory Factor Analysis. *Bayesian Analysis*. <https://doi.org/10.1214/26-BA1611>
2. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J. & Vadillo, M. A. (2026). A Unified Framework for Psychometrics in Experimental Psychology: The Standardized Generalized Hierarchical Factor Model. *Psychological Methods*. Advance online publication. <https://doi.org/10.1037/met0000846>
3. Garre-Frutos, F., Martínez-López, P., Ariza, A., **Rey-Sáez, R.**, Luque, D., Vadillo, M. A., & González, F. (2026, March 22). Explicit knowledge modulates Pavlovian attentional biases. *PsyArxiv*. https://doi.org/10.31234/osf.io/uvge3_v2
4. Franco-Martínez, A., **Rey-Sáez, R.**, Shanks, D. R., Soto, D., & Vadillo, M. A. (2026). Replicating the unconscious working memory effect: A multisite preregistered study. *Neuroscience of Consciousness*. <https://doi.org/10.1093/nc/niaf046>
5. Pacheco-Romero, A. M., Martín-García, Ó., **Rey-Sáez, R.**, Boemo, T., Blanco, I., Vázquez, C., & Sánchez-López, Á. (2024). An integrative analysis of potential mechanisms of reduced positive affect in daily life in depression: an ESM study. *Cognition and Emotion*, 38(4), 587–604. <https://doi.org/10.1080/02699931.2024.2314066>
6. Gracia, E., Martín-Fernández, M., **Rey-Sáez, R.**, & Yount, K. M. (2024). Attitudes justifying partner violence against women in Latin-American and Caribbean countries: A measurement invariance study. *Psychology of Violence*, 14(2), 127–136. <https://doi.org/10.1037/vio0000497>
7. Vega-Sanz, M., **Rey-Sáez, R.**, Berástegui, A., & Sánchez-López, Á. (2024). Difficulties in pregnancy adjustment as predictors of perinatal depression: Indirect effects through the role of brooding rumination and maternal-fetal bonding. *American Journal of Perinatology*, 41(S 01), e2870–e2877. <https://doi.org/10.1055/s-0043-1776062>
8. Franco-Martínez, A., **Rey-Sáez, R.**, & Castillejo, I. (2023). The seven wonderings of large language models as psychometric designers, refiners, and analysts. *PsyArXiv*. <https://doi.org/10.31234/osf.io/kmqy5>
9. **Rey-Sáez, R.** (2022). Réplica a: Validación cruzada sobre una misma muestra: Una práctica sin fundamento. *R.E.M.A. Revista Electrónica de Metodología Aplicada*, 24(1), 41–44. <https://doi.org/10.17811/rema.24.1.2022.41-44>

Conference Presentations

1. **Rey-Sáez, R.** (2026). *Experimental Psychometrics*. [Symposium Organizer] XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS).
2. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J., & Vadillo, M. A. (2026). *Don't Aggregate-then-Correlate, Model! Unleashing the Power of Generalized Hierarchical Factor Models in Experimental Psychology*. [Oral presentation] Symposium: *Experimental Psychometrics*. XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS).
3. Garre-Frutos, F., Franco-Martínez, A., Vadillo, M. A., & **Rey-Sáez, R.** (2026). *A multilevel-signal detection framework for unconscious processing research*. [Oral presentation] Symposium: *Experimental Psychometrics*. XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS).
4. Garre-Frutos, F., Martínez-López, P., **Rey-Sáez, R.**, Luque, D., Vadillo, M. A., & González, F. (2026). *Explicit knowledge modu-*

- lates Pavlovian attentional biases*. [Oral presentation] Symposium: *Current Debates in Research on Habits*. XV Conference of the Spanish Society for Experimental Psychology (SEPEX) and II Joint Meeting of the SEPEX and the Belgian Psychological Association (BAPS).
5. Franco-Martínez, A., **Rey-Sáez, R.**, Revuelta, J., & Vadillo, M. A. (2026). *Cuando el análisis factorial empezó a contestar preguntas de psicología experimental*. [Oral presentation] Symposium: *Nuevos desarrollos en Análisis Factorial*. XIX Congreso de Metodología de las Ciencias Sociales y de la Salud.
 6. **Rey-Sáez, R.**, & Revuelta, J. (2026). *Rotaciones Oblicuas en AFE Bayesiano: superando a los métodos frecuentistas en su propio terreno*. [Oral presentation] Symposium: *Nuevos desarrollos en Análisis Factorial*. XIX Congreso de Metodología de las Ciencias Sociales y de la Salud.
 7. **Rey-Sáez, R.** (2026). *Simultaneous Latent Dimensionality Assessment and Factor Loading Estimation in Sparse Bayesian Exploratory Factor Analysis*. [Oral presentation] XIX Congreso de Metodología de las Ciencias Sociales y de la Salud.
 8. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J., & Vadillo, M. A. (2026). *A Multilevel Psychometric Framework for Experimental Tasks: The Generalized Hierarchical Factor Model* [Oral presentation] 15th International Multilevel Conference.
 9. Franco-Martínez, A., **Rey-Sáez, R.** & Vadillo, M. A. (2026) *How do I start planning my multisite project? A tutorial on multilevel simulation-based power and sensitivity analysis* [Oral presentation] 15th International Multilevel Conference.
 10. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J., & Vadillo, M. A. (2025). *New Hierarchical Models to Answer Old Psychometric Questions in Experimental Research: Which Task Best Measures Inhibition?* [Oral presentation] XIV Conference – Asociación Iberoamericana para la Investigación de las Diferencias Individuales (AIIDI).
 11. **Rey-Sáez, R.**, Franco-Martínez, A., Revuelta, J., & Vadillo, M. A. (2025). *Where psychometrics meets experimental psychology: Bayesian hierarchical factor models for response times*. [Oral presentation] XI Conference – European Congress of Methodology.
 12. **Rey-Sáez, R.**, Revuelta, J., & Vadillo, M. A. (2025). *From random effects to common factors: Latent dimensionality assessment in experimental psychology*. [Oral presentation] XI Conference – European Congress.
 13. **Rey-Sáez, R.**, Garre-Frutos, F., Franco-Martínez, A., Castillejo, I., & Vadillo, M. A. (2025). *Addressing methodological challenges in unconscious process research: A hierarchical modeling approach*. [Poster presentation] 27 Annual Meeting of the Association for Scientific Studies of Consciousness.
 14. **Rey-Sáez, R.**, Garre-Frutos, F., Franco-Martínez, A., Castillejo, I., & Vadillo, M. A. (2025). *Hierarchical models in experimental psychology: A solution for measuring unconscious psychological processes*. [Poster presentation] XIV Reunión Científica sobre Atención.
 15. **Rey-Sáez, R.**, Garre-Frutos, F., & Vadillo, M. Ángel. (2024). *What psychometrics can offer to experimental psychology: The multilevel model as the legacy of factor analysis*. [Oral presentation] XVIII Congreso de Metodología de las Ciencias Sociales y de la Salud.
 16. **Rey-Sáez, R.**, Lecuona, Ó., & Martín-Fernández, M. (2024). *Approximate measurement invariance with multiple groups: A comparison between Bayesian SEM and the alignment method*. [Oral presentation] XVIII Congreso de Metodología de las Ciencias Sociales y de la Salud.
 17. Lecuona, Ó., **Rey-Sáez, R.**, & Martín-Fernández, Manuel. (2024). *Studying invariance with multiple groups in large datasets: A comparison of bayesian SEM and alignment method*. [Oral presentation] 13th Conference of The International Test Commission.
 18. **Rey-Sáez, R.**, Franco-Martínez, A., Castillejo, I., & Vadillo, M. A. (2024). *From unconscious to unreliable? Measurement error biases evidence on studies of unconscious processing*. [Poster presentation] 27 Annual Meeting of the Association for Scientific Studies of Consciousness.
 19. Franco-Martínez, A., **Rey-Sáez, R.**, Shanks, D. R., Soto, D., & Vadillo, M. A. (2024). *Advances in “replicating the unconscious working memory effect: A multisite preregistered study.”* [Poster presentation] 27 Annual Meeting of the Association for Scientific Studies of Consciousness.
 20. Pacheco-Romero, A. M., Martín-Gracia, Ó., **Rey-Sáez, R.**, Boemo, T., Blanco, I., & Sánchez-López, Á. (2023). *Affective reactivity to daily events predicts changes in future psychopathology and is modulated by the use of different forms of rumination*. [Oral presentation] International Convention of Psychological Science (ICPS).
 21. Aivar, P., **Rey-Sáez, R.**, & Vadillo, M. A. (2023). *Hard to ignore? Irrelevant distractors cannot be completely suppressed in a contextual cueing task*. [Poster presentation] 23 Annual Meeting of the Vision Sciences Society.
 22. Martín-García, O., **Rey-Sáez, R.**, Boemo, T., Pacheco-Romero, A., Blanco, I., & Sánchez-López, A. (2023). *Modulatory role of motivational self-focuses on the relation between perceived stress and momentary affect: An ESM study*. [Oral presentation] Emotions Congress 2023.
 23. Pacheco-Romero, A. M., Martín-Gracia, Ó., **Rey-Sáez, R.**, Boemo, T., Blanco, I., Vázquez, C., & Sánchez-López, Á. (2022). *Intra-individual mechanisms of positive affect regulation deficits in depression: An experience sampling study*. [Oral presentation] 52nd European Association for Behavioural; Cognitive Therapies.
 24. **Rey-Sáez, R.**, & Nájera, P. (2022). *Investigando metodología en el multiverso de la academia: Comunidad, recursos y estrate-*

- gias por y para jóvenes. [Symposium Organizer] XVII Congreso de Metodología de las Ciencias Sociales y de la Salud.
25. **Rey-Sáez, R.** (2022). *Investigación, paso 1: ¿Cómo y dónde busco información? Nuevas herramientas para la detección eficiente de bibliografía de interés.* [Oral presentation] XVII Congreso de Metodología de las Ciencias Sociales y de la Salud.
 26. **Rey-Sáez, R.** (2022). *Approximate measurement invariance: Assessing the accuracy and sensitivity of alignment optimization under partial invariance with multiple groups.* [Oral presentation] XVII Congreso de Metodología de las Ciencias Sociales y de la Salud.
 27. Martín-Fernández, M., & **Rey-Sáez, R.** (2022). *Approximate measurement invariance in surveys on attitudes toward intimate partner violence in 12 Latin American countries.* [Poster presentation] XVII Congreso de Metodología de las Ciencias Sociales y de la Salud.
 28. **Rey-Sáez, R.** (2022). *Approximate measurement invariance in surveys on attitudes toward intimate partner violence in 12 Latin American countries.* [Oral presentation] Research Conference of the Faculty of Psychology, Universidad Autónoma de Madrid